WBMME – Final Project Report and Deviation Log

User Story: US-36 – Final Report and Deviation Log
Project: Web-Based ML Model Evaluator (WBMME)
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1. Executive Summary

The Web-Based ML Model Evaluator (WBMME) is a Django-based web application that supports machine-learning workflows, including dataset management, preprocessing, model training, evaluation, experiment tracking and comparison.

The application also provides authentication, role-based access control and administrative functionality.

The latest full automated test execution completed with **983 passed, 1 skipped, 2 expected failures and 0 failed tests**. The application achieved **99% code coverage and 96.84% branch coverage**.

The previously intermittent SQLite locking issue in the live-server training benchmark was addressed by configuring a file-backed SQLite test database. The benchmark passed three consecutive targeted runs and also passed in the full test suite.

2. Main Application Modules

Module	**Functionality**
Accounts	Registration, login, authentication, roles and permissions
Datasets	Dataset upload, validation, preview, profiling and preprocessing
Training	Model catalogue, training configuration, execution and run status
Evaluation	Evaluation metrics, confusion matrix and feature importance
Experiments	Experiment history, run comparison and export
Audit	Activity tracking and audit records
Administration	Administrative user and dataset management

3. User Roles

Student: Provides a learning-oriented interface with a separate Models navigation option.

Data Scientist: Provides an advanced dashboard. Model-related activities are accessed through the training and experiment workflow rather than a separate Models menu item.

The distinction is primarily in the interface and workflow presentation. It should not be represented as a difference in model capabilities unless enforced by the application's permissions.

4. Testing Approach

The project uses automated testing to verify:

- Authentication and role-based access.
- Dataset upload, validation and preprocessing.
- Model configuration and training.
- Evaluation metrics and visualizations.
- Experiment history and comparison.
- Report export.
- Administrative and audit functionality.
- Security, performance and reliability.

The test suite includes unit, integration, system and regression tests.

5. Final Test Execution Results

Metric	**Result**
Total tests	986
Passed	983
Failed	0
Skipped	1
Expected failures (XFAIL)	2
Execution time	26 min 23 sec

Result: The full test suite completed with zero failures.

System Test Results

The Appendix A system test execution produced:

- 9 passed
- 1 expected failure
- 0 failed

The expected failure relates to the post-login navigation observation described in the deviation log.

6. Code Coverage and Branch Coverage

Coverage metric	**Achieved**
Code coverage (statement/line coverage)	**99%**

Branch coverage	**96.84%**

Coverage Details

Measurement	Total	Covered	Missing
Statements	6,217	6,154	63
Branches	632	612	20

Code coverage measures the proportion of executable statements exercised by the test suite.

Branch coverage measures the proportion of decision branches exercised by the test suite.

These are separate coverage measurements.

7. Performance and Stability

The live-server training benchmark initially experienced intermittent SQLite table-locking errors when the test server and background training thread accessed a shared in-memory test database.

The test database configuration was changed to use a file-backed SQLite database.

The benchmark was then executed three consecutive times:

Run	Result	Execution time
Run 1	Passed	36.73 sec
Run 2	Passed	35.54 sec
Run 3	Passed	35.49 sec

The benchmark also passed during the latest full test-suite execution.

8. Deviation and Observation Log

ID	Area	Observation / deviation	Status
O-5	Login navigation	After successful login, the application redirects to the Welcome page instead of directly to the Dashboard.	Open / documented
D-1	Dataset split validation	A boundary value of 0.151 is rounded and accepted as 15% under the current validation behavior.	Verify against the latest test output
OBS-03	Login performance	A burst-login test recorded approximately 2.4 seconds average and 3.1 seconds p95.	Performance observation
OBS-04	SQLite test database	Shared in-memory SQLite caused intermittent locking in the live-server benchmark. The file-backed test database configuration resolved the observed issue in three consecutive targeted runs and the full suite.	Mitigated and retested

VERIFY-01	Test execution	One test was skipped and two tests were marked as expected failures in the latest full run.	Record exact reasons from pytest output

Expected failures and skipped tests are documented rather than counted as successful passes or silently omitted.

9. Demonstration Checklist

The following workflows should be demonstrated:

1. Registration and login.
2. Student dashboard and Models navigation.
3. Data Scientist advanced dashboard.
4. Dataset upload and management.
5. Dataset preview and preprocessing.
6. Model training configuration and execution.
7. Evaluation metrics and visualizations.
8. Experiment history and run comparison.
9. Report export.
10. Administrative functionality and audit records.
11. Docker-based application execution.

10. Screenshot and Supporting Evidence Register

Insert genuine screenshots captured from the final running application.

ID	Required evidence
SS-01	Login and registration
SS-02	Student dashboard
SS-03	Data Scientist advanced dashboard
SS-04	Dataset upload and listing
SS-05	Dataset preview and preprocessing
SS-06	Model/training configuration
SS-07	Completed training run
SS-08	Evaluation results
SS-09	Experiment history and comparison

SS-10 Export and administration

SS-11 Docker execution

SS-12 Successful GitHub Actions CI run

SS-13 Final pytest and coverage summary

The following project documents should also be attached:

- Software Requirements Specification (SRS).
- Software Architecture Document (SAD).
- Requirements Traceability Matrix (RTM).
- Software Test Plan and Appendix A.
- Final test execution output.
- Coverage report.

11. Development Freeze

Before declaring the development freeze complete:

- Final test results and coverage evidence saved.
- Known deviations documented and reviewed.
- SRS, SAD, RTM and Test Plan attached.
- Genuine application screenshots attached.
- Latest CI status and commit SHA recorded.
- Git working tree checked for uncommitted changes.
- Credentials, local database files and demo_logins.txt excluded from commits.
- No further feature development after the approved final baseline; only necessary critical fixes permitted.

12. Conclusion

The latest WBMME test execution completed with **983 passing tests and zero failures**.

The application achieved **99% code coverage and 96.84% branch coverage**. The live-server training benchmark passed three consecutive targeted runs following the file-backed SQLite test database configuration.

Known deviations and test observations have been documented for project evaluation and demonstration.

The final evidence pack must include the approved baseline documents, genuine application screenshots, exact skipped/XFAIL explanations and repository freeze evidence before the US-36 acceptance criteria can be declared fully satisfied.